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PROCEDURE

- Set up the sonometer on the table and clean the groove on the pulley to ensure minimum friction
- Stretch the wire by placing suitable mass in the hanger
- Set the tuning fork into vibrations by striking it against the rubber pad. Plug the sonometer wire and compare the two sounds.
- Adjust the vibrating length of the wire by sliding the bridge B till the sounds appear alike.
- For the final adjustment, place a small paper rider R in the middle of the wire AB.
- Sound the tuning fork and place its shank stem on the bridge A or on the sonometer box and slowly adjust the position of bridge B until the paper rider is agitated violently indicating resonance.
- The length of the wire between the wedges A and B is measured using meter scale. It is the resonant length. Now the frequency of vibration of the fundamental mode equals the frequency of the tuning fork.
- Repeat the above procedure for other tuning forks by keeping the same load in the hanger.

OBSERVATIONS

Tension (constant) on the wire (mass suspended from the hanger including its own mass)
T = _____ N

Variation of frequency with length		
Frequency of the tuning fork 'n ' (Hz)	Resonant length *'l'*	*nl*
n_1 =		
n_2 =		
n_3 =		
n_4 =		
n_5 =		
n_6 =		